

Wyatt Walsh

SENIOR AI/ML ENGINEER

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Summary: Senior AI/ML engineer building production agentic AI, LLM document intelligence, RAG, and data platforms for fintech, risk, compliance, and enterprise tooling.

EXPERIENCE

Senior AI/ML Engineer

Frame Payments | Remote | Oct 2025 - Jan 2026

Delivered production agentic AI document intelligence, human-in-the-loop review, and multimodal compliance screening for high risk merchant onboarding.

- Deployed production LLM document intelligence with PydanticAI, Amazon Bedrock AgentCore, and Claude; added human-in-the-loop browser feedback for confidential uploads and reduced analyst follow-ups/resubmissions ~90% by blocking miscategorized documents before review.
- Built multimodal restricted-item compliance screening by combining product images and structured metadata; translated banking-partner policy requirements into automated risk checks for merchant onboarding across tens of merchants and multi-location catalogs.
- Built a NotebookLM-based technical knowledge hub spanning vendor/web documentation and internal monorepo docs; automated corpus refresh with a Colab pipeline that re-ingests the latest sources to keep retrieval accurate and up to date.
- Drove internal AI adoption with Notion MCP playbooks, high-signal prompt patterns, and weekly office hours reaching ~15% of the company; standardized PII/confidential-document safeguards and auditability requirements.

Associate III, Senior Software Engineer - Data Integrations

JPMorgan Chase & Co. | New York, NY | May 2023 - Apr 2025

Built enterprise data pipelines, risk analytics, and data-quality automation for credit trading while scaling secure internal AI developer tooling.

- Pioneered firm-wide rollout of secure generative AI developer tools (LLMs, GitHub Copilot, Codeium) as one of the first 500 engineers, shaping scalable deployment and leading workshops for technical and non-technical teams.
- Devised ML-driven data-quality anomaly detection for credit trading and risk analytics, reducing false positives by 25% and cutting manual review time so risk teams could focus on critical alerts.
- Developed resilient Python wrapper for KDB+, democratizing data access for 5,000+ users with CI-style tests and optimized Q functions, reducing manual processing by over 10 hours per week.
- Engineered a cloud-function-driven asset-fee data sink, consolidating 10+ sources into a centralized database, cutting manual maintenance by 90% and providing on-demand access for traders through a Python SDK, REST API, and interactive dashboard.

Senior Software Engineer

Stealth Web3 Startup | Remote | Mar 2022 - Jul 2022

- Architected a scalable NFT-collateralized lending DApp using React, Next.js, TypeScript, MobX, and ethers.js, enabling secure, performant on-chain loans.
- Implemented end-to-end CI/CD with GitHub Actions, Husky, Cypress, and Jest to halve release cycles and raise test coverage to 90%, while mentoring a front-end intern to build key UI components.
- Optimized front-end performance through code splitting, lazy loading, and asset caching, reducing initial load times by 40% and JavaScript bundle size by 30%.

Founder, Chief Technology Officer

SandLabs | Remote | Jun 2021 - Mar 2022

- Engineered and launched a Next.js/TailwindCSS marketing site with top-tier Lighthouse scores and secured \$75K in OCEAN tokens via a competitive DAO grant.
- Engineered Python pipelines to extract and normalize a 1M+ Web3-related Reddit posts & comments dataset and published the data on Ocean Protocol's Data Marketplace.

Demand Forecasting Extern

Gap Inc. | San Francisco, CA | Jan 2020 - May 2020

- Applied time-series decomposition and bias correction techniques to demand forecasts, boosting accuracy by up to 25% and minimizing surplus inventory costs.
- Owned the data engineering layer by architecting scalable MySQL and Apache Spark pipelines across 5+ years of sales data, improving data processing throughput by 95% and standardizing insights across forecasting teams.

SKILLS

MACHINE LEARNING AND GENERATIVE AI: scikit-learn, Hugging Face Transformers, OpenAI API, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Prompt Engineering, PydanticAI, Amazon Bedrock, Claude, Claude Code, GitHub Copilot, DeepAgents, AI Agent Skills Protocol

DATA ENGINEERING AND ANALYTICS: Pandas, NumPy, SQLAlchemy, Pydantic, Pandera, Apache Spark, Amazon Kinesis, AWS Glue, KDB+, PostgreSQL, DynamoDB, ETL, SQLite

PROGRAMMING & SCRIPTING LANGUAGES: Python, SQL, TypeScript, JavaScript, Go, Rust, Bash

CLOUD INFRASTRUCTURE AND DEVOPS: AWS, Docker, Kubernetes, Terraform, GitHub Actions, Serverless Architecture, AWS Lambda, Google Cloud Functions, ECS, EKS, Prometheus, Grafana

PROJECTS

AI Agent Harness Configs | github.com/wyattowalsh/agents

LLM developer-workflow catalog with reusable skills, MCP-ready configs, and cross-runtime agent automation.

- Built 40+ reusable instruction and skill packs plus shared harness configs that install across Claude Code, GitHub Copilot, Cursor, and related agent runtimes.
- Shipped generated docs, install flows, and portable tool and MCP presets for code review, security, infrastructure, research, and workflow automation.

Tech: Python, TypeScript, MCP, FastMCP

NBA Basketball Database | github.com/wyattowalsh/nbadb

Validated data warehouse spanning 65K+ NBA games and 13M+ play-by-play events with ETL, docs, CLI, and Kaggle distribution.

- Implemented a high-performance, validated ETL pipeline that extracts from the NBA API, updates warehouse outputs daily, and exports DuckDB, SQLite, Parquet, and CSV artifacts.
- Built the surrounding product surface with docs, schema references, release workflows, and Kaggle distribution that pushed the dataset past 400K views and 45K downloads.

Tech: Python, Pandas, SQLAlchemy, SQLite, Kaggle

ProxyWhirl | github.com/wyattowalsh/proxywhirl

Data-collection toolkit with nine rotation strategies, 100+ auto-validated sources, and self-healing proxy health checks.

- Implemented 9 rotation strategies (round-robin, weighted, geo-targeted, performance-based, and more) with configurable fallback chains and per-source health scoring.
- Built a self-healing pipeline that continuously validates 100+ built-in proxy sources, auto-ejects unresponsive proxies via circuit breakers, and re-promotes recovered sources.

Tech: Python, SQLite, PyPI

AI/ML Web Feeds | github.com/wyattowalsh/ai-web-feeds

Typed AI/ML source-discovery platform with validation, contributor workflows, and docs-backed feeds for model and agent research.

- Defined a structured YAML feed schema with topic classification (ML, NLP, CV, etc.) and source-type validation (blog, paper, newsletter) to ensure feed quality at scale.
- Shipped helper scripts, CI checks, scheduled validation jobs, and a live docs surface so community-submitted feeds can be reviewed, normalized, and published reliably.

Tech: Python, YAML, Vercel

MCP-Crawl4AI | github.com/wyattowalsh/mcp-crawl4ai

FastMCP server enabling AI agents to crawl, extract, and return structured web content over documented stdio/HTTP paths.

- Built a FastMCP v3 server with lifespan-managed headless Chromium, session-aware crawling workflows, and automatic Playwright browser setup—only 2 runtime dependencies.
- Exposed 4 tools, 2 resources, and 3 prompt templates with both stdio and HTTP transport, backed by docs, tests, and release-tagged install flows for MCP-compatible agents.

Tech: Python, FastMCP, Crawl4AI, Playwright, Docker

MDXPad | github.com/wyattowalsh/mdxpad

Native macOS MDX editor with synchronized preview, multi-provider AI assistance, and Keychain-backed credential storage.

- Built a dual-format editor on Electron 39 + CodeMirror 6 with synchronized scroll between source and rendered preview, remark/rehype plugin support, and YAML frontmatter validation.
- Integrated multi-provider AI assistance with bring-your-own-key support, per-provider token/cost tracking, and Keychain-backed secret storage.

Tech: Electron, React, TypeScript, CodeMirror, Vercel AI SDK

Personal Website: w4w.dev | github.com/wyattowalsh/personal-website

Long-running personal web app and MDX publishing system pairing a polished frontend with durable content, SEO, and authoring workflows.

- Designed and maintained the public product surface across multiple generations, including responsive landing experiences, theme support, motion, and accessibility-conscious UI polish.
- Built a deeply customized MDX publishing workflow with search, metadata, math rendering, feeds, tags, and other authoring features usually split across several separate tools.

Tech: Next.js, React, TypeScript, Tailwind CSS, MDX

EDUCATION & CERTIFICATIONS

Bachelor of Science

University of California, Berkeley - College of Engineering, Industrial Engineering & Operations Research

Aug 2014 - Dec 2020

High School Diploma

The Hotchkiss School

Aug 2012 - May 2014

CERTIFICATIONS: AWS Certified Cloud Practitioner | IBM Data Science Professional Certificate